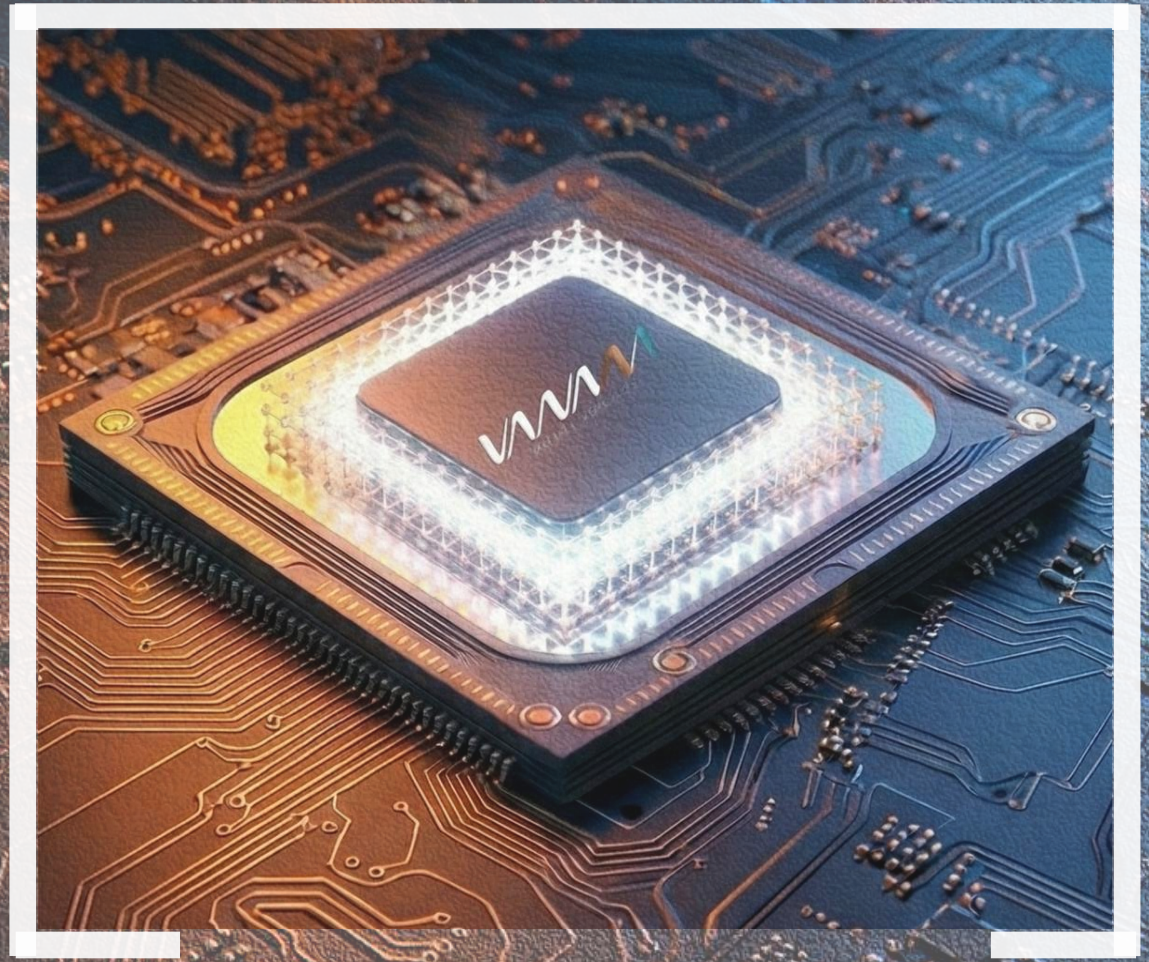


# Next-Generation Foundry for Quantum Photonics

VANA MATERIALS LAB.

# Company Overview

> Founded in 2022, dense micro-foundry foundation in Seoul, current strategic focus on quantum photonics.

## COMPANY FACTS

| | |
|---|---|
| Founded | Jan. 27, 2022 |
| 1st Lab | Yeongdeungpo, Seoul |
| Fab footprint | ~1,600 ft$^2$ |
| Team | 13 employees |
| Management | PhD-led |
| Customers | 42 paying |
| IP | 23 patents + 3 TMs |
| Funding | Seed & $3M+ non-dilutive |

## OPERATING CAPABILITY

1. **Deposition**
   Cluster Sputter/ Inline Sputter/ ALD / Evaporator
2. **Metrology**
   Ellipsometer / Parameter analyzer
3. **Back-end**
   Laser dicing; external packaging and cryogenic test network
4. **Materials**
   ~40 oxides, nitrides and metals; quantum focus on NbTiN, $Ta_2O5$, BTO, STO and DBR stacks

## STRATEGIC EVOLUTION

- **2022** — Company founded; Seoul micro-foundry initiated
- **2023-25** — Seed funding, R&D grants, customer projects, portfolio and IP build-out
- **2025-26** — U.S. market exposure through Luminate Accelerator and U.S. events
- **2026-28** — **Focused build of U.S. quantum photonics foundry**

**NEW POSITIONING: SPECIALTY QUANTUM PHOTONICS FOUNDRY**

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# The Era's Bottleneck: Software Outpacing Hardware

**AI compute demand exploding**

**Hardware reaching physical/energy limits**

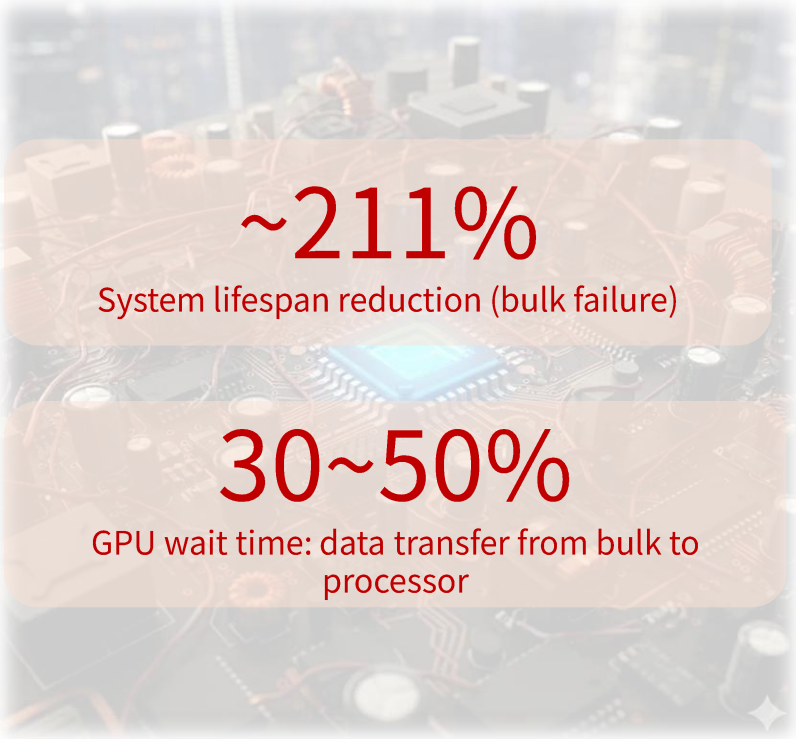



**21st Century Innovation** VS **20th Century Hardware**

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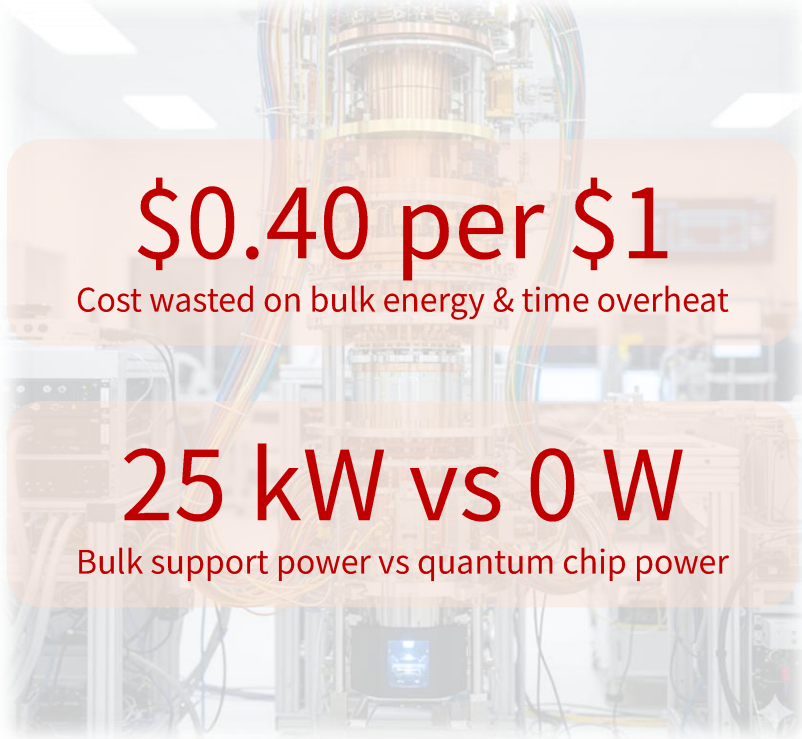
# The Invisible Barrier of The AI Generation

> Innovation: currently bottlenecked by inefficient hardware & traditional bulk components.



**~211%**
System lifespan reduction (bulk failure)

**30~50%**
GPU wait time: data transfer from bulk to processor



**$0.40 per $1**
Cost wasted on bulk energy & time overheat

**25 kW vs 0 W**
Bulk support power vs quantum chip power

**System failure caused by traditional bulk components: ~211% of system lifespan reduction (in automotive), up to 50% of wasted time (in AI systems), intense power usage by bulk/support system as opposed to the actual quantum chip (in quantum systems).**

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# Problem: Traditional Foundries & The Need for New Materials

> Quantum photonics performance determined by thin film quality, interface & device fabrication, long before software.

### Quantum System Requirement

- Single-photon detection
- Low-loss optical routing
- Cryogenic modulation
- High-reflectivity mirrors
- Integrated photonic circuits

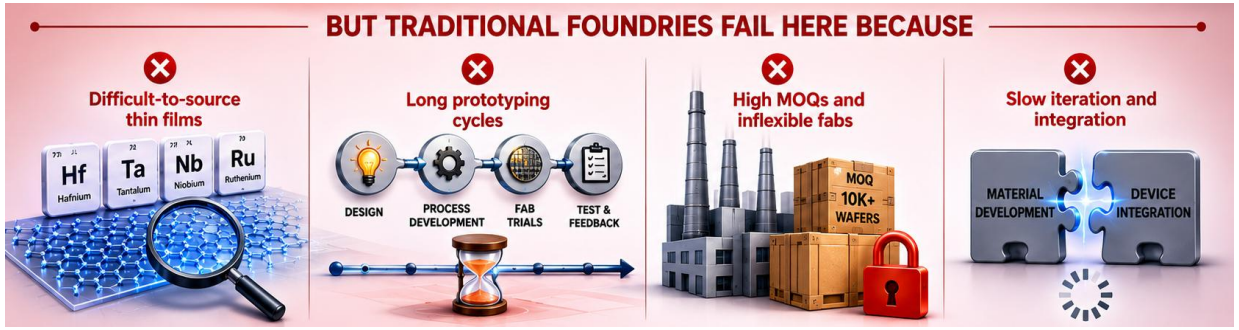
### Thin-Film / Process Variables

- Stoichiometry and crystallinity
- Thickness and wafer uniformity
- Surface / interface roughness
- Film stress and adhesion
- Pattern transfer / critical dimension
- Cryogenic electrical and optical behavior

### Commercial Failure Mode

- Device performance misses specification
- Iteration cycles extend by months
- Multiple vendors create handoff loss
- Qualification data is not repeatable
- Customer engineering teams build ad hoc internal processes




**Software and system design cannot compensate for unstable material and process foundations.**

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# Customer Pain Points

> Time & capital loss for early quantum hardware teams due to vendor-fragmented deposition, nano-patterning, metrology & cryogenic validation.

**MATERIAL ACCESS**

*"We cannot source the film at the required stoichiometry and thickness."*

**PROCESS FLEXIBILITY**

*"The foundry will not alter its standard process stack."*

**LOT ECONOMICS**

*"We need 5-50 wafers, not a volume contract."*

**VENDOR HANDOFF**

*"Deposition and nano-patterning sit with different vendors."*

**LATE VALIDATION**

*"Cryogenic failure is discovered after the fabrication cycle."*

**IP / DATA**

*"Recipe data and process IP ownership are unclear."*

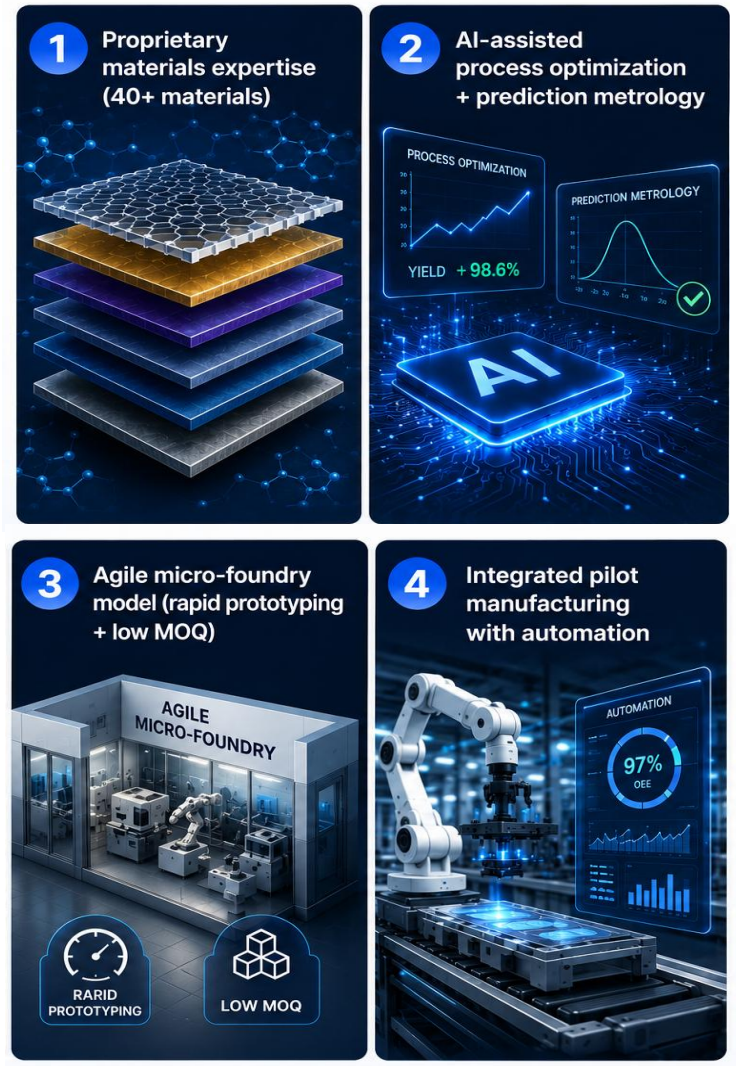
**Impact: longer qualification cycles | duplicated setup cost | uncertain yield | diversion of scarce quantum engineeringtalent**

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# Solution: VanaM Specialty Quantum Photonics Foundry



> Patented, AI-assisted process platform for next-generation quantum hardware commercialization.




- Custom materials & device iteration before scale.
- Customer-specific oxides, nitrides, metals & multilayers.
- Low-volume, high-value dedicated runs.
- Recipe, process-window & device co-development options.
- Custom nanowire/photonic structures within tool capacity.
- DOE-based recipe & device co-development.
- Partnered cryogenic test & package path.

**21st Century Solution for 21st Century Innovation**

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# Solution: Quantum Photonics Foundry-Platform

> VanaM: one commercial workflow for custom material deposition + nano-patterning device fabrication + qualification data + partner-enabled packaging.

## 1 Materials Foundry

Custom oxides, nitrides, metals and multilayer stacks; recipe development and low-volume wafers.

## 2 Device Foundry

Nano-patterned SNSPD, waveguide, modulator, DBR and PIC structures within qualified process windows.

**VanaM**
**QUANTUM PHOTONICS FOUNDRY**

## 3 Process Qualification

DOE, metrology, run-to-run data, specification control and recipe transfer from Korea to the U.S.

## 4 Integration Network

Dicing, packaging, fiber coupling and cryogenic test through internal capability and qualified partners.

NRE / DOE → Pilot Wafer → Patterned Device → Repeat Lot → Package / Test → IP / JD

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# Business Model

> Value before scale-up, then wallet expansion as customers move from samples to wafers, devices, packaged components & PIC supply.

### 1 Process NRE

DOE, recipe setup, engineering and qualification milestones

### 2 Material Wafers

Korea-made thin-film wafers: NbTiN, Ta2O5, BTO/STO and DBR stacks.

### 3 Patterned Devices

U.S. lithography, nano-patterning, etch, release and screened die sales.

### 4 Package / Test

Dicing, fiber/electrical package and cryogenic performance verification.

### 5 Integrated Quantum PIC

1Q-eq evaluation PIC, 8Q module and 32Q tile supply programs.

### 6 IP / Joint Development

License, royalty, exclusivity or co-development around platform processes.

## CUSTOMER MATURATION AND WALLET EXPANSION

**Sample / Feasibility**
**$10k-$50k**
Material sample, characterization or limited DOE.

>

**Paid Qualification**
**$50k-$250k**
Process development and pilot wafers.

>

**Material / Device Supply**
**$0.3M-$1.5M**
Recurring wafers, dies, packages and engineering support.

>

**Integrated PIC Eval Lot**
**$0.5M-$2.0M**
10-50 1Q-eq PICs, 8Q modules and qualification support.

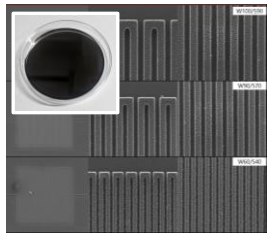
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**Strategic PIC Supply**
**$2M-$10M+**
Multi-tile supply, platform JD, exclusivity or annual framework.

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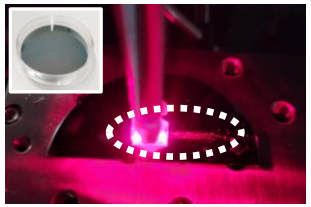
# Quantum Photonics Project

## SNSPD



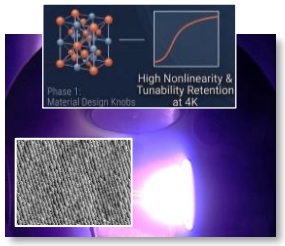
High-sensitivity superconductors

## Waveguide



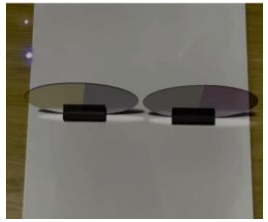
Low loss optical waveguides

## Modulator



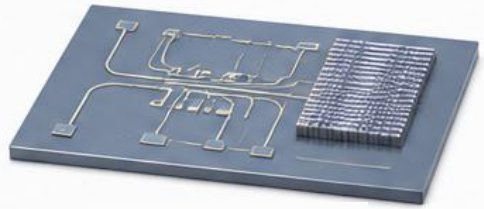

High cryogenic electro-optic modulation

## DBR



High-reflection DBR wafer

## Photonics Integrated Circuits (PIC)



## Key Thin Films Next-Gen PICs & for Quantum Systems

- ✓ **SNSPD: NbTiN** ⇢ Superconducting transition T ~ 11K
- ✓ **Waveguide: $Ta_2O_5$** ⇢ Propagation loss < 0.5 dB
- ✓ **Modulator: BTO, STO** ⇢ Pokel coefficient ~1000pm/V at 5K
- ✓ **DBR: Si-$SiO_2$-$TiO_2$** ⇢ Reflectance > 99%

KIST Quantum Fab & UIUC Collaboration

## Scalable Opportunities / Areas

| Quantum Computers | Quantum Communication | Quantum Sensors | Laser & LiDAR | Glass Core Substrate | Silicon Photonics |
|---|---|---|---|---|---|
| $13 Bn | $12 Bn | $1.6 Bn | $22 Bn | $223 Bn | $1.9 Bn |
| CAGR 32% | CAGR 30% | CAGR 26% | CAGR 30% | CAGR 8% | CAGR 17% |

Source: MarketsandMarkets, MMR, Precedence Research, Fact.MR

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# Market Expansion Opportunities

> Bottom-up account investable market case & wallet-share opportunity within a much larger, fast-growing quantum and advanced photonics ecosystem.

## MARKET SIZE VALUE

| | | |
|---|---|---|
| $60B By 2030 | TAM | Quantum computing, communication, sensing, silicon photonics, LiDAR, advanced photonics |
| $4B By 2030 | SAM | Custom quantum photonics thin films, wafers, devices and related process services |
| $400M By 2030 | BEACHHEAD | U.S. low-volume, high-value customers requiring custom wafers and nano-patterned devices |

**Growth signal** *McKinsey: quantum communication **~22-25% CAGR** through 2035; core quantum technology revenue up to **$97B** by 2035.*




*Sources: Grand View Research, MarketsandMarkets, Yole Group, KBV Research, 2025–2026 industry reports.*

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# Traction & Validation

> U.S. market entry with operating KR fab, growing repeat-customer revenue base, 50 material process library, non-dilutive funding & recognized U.S. photonics ecosystem access.

## KOREA CUSTOMER BASE QUALITY (FY ~26)

**42** **Active accounts**
cumulative 42 paying customers to date

| | |
|---|---|
| Active accounts | 42 |
| Repeat accounts (2+ orders) | 29 |
| Repeat-order share of revenue | 91.3% |
| Average annual value per Repeat accounts | $8.6K |
| '23 ~ '25 Active accounts CAGR | 169.3% |
| University : R8D Institute : Corporate | 55 : 25 : 20 |

## MATERIALS & PROCESS LIBRARY (49 FILMS)

**50** **Thin-film materials — process secured**
special films & process windows beyond conventional fabs



## ECOSYSTEM ANCHOR POINTS (public)

**KIST Quantum Fab · UIUC**
Collaboration pathways for cryogenic and device validation

**Intelon Optics (Boston, MA)**
Spectrometer PIC & CHIPS act grant proposal partner

**Stratio Inc. (San Jose, CA)**
Photonics-sensor ecosystem partner

**NextCorps Luminate Cohort 8 (Rochester, NY)**
AIM Photonics · Thorlabs · Edmund Optics · Omega Optical network touchpoints

**SelectUSA Investment Summit 2026 — 1st place, Open Tech Pitching (2025: 2nd)**
Repeat top ranking before U.S. investors and EDOs

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# U.S. Expansion – Strategy

> A new U.S entity to build and operate Phase 1 Foundry, KR-US dual-site model.

## KOREA FOUNDRY

**PROCESS DEPTH + REVENUE**

- Existing customer production
- Material recipe development
- Cross-site transfer standards
- Metrology and technical support
- Training and troubleshooting

## U.S. PHASE 1 FOUNDRY

**8-INCH + NANO-PATTERNING**

- Customer-facing process qualification
- 8-inch deposition and multilayer stacks
- Nano-patterned quantum photonic devices
- U.S. technical team and program access
- Domestic supply-chain positioning

## PARTNER NETWORK

**EXTEND THE VALUE CHAIN**

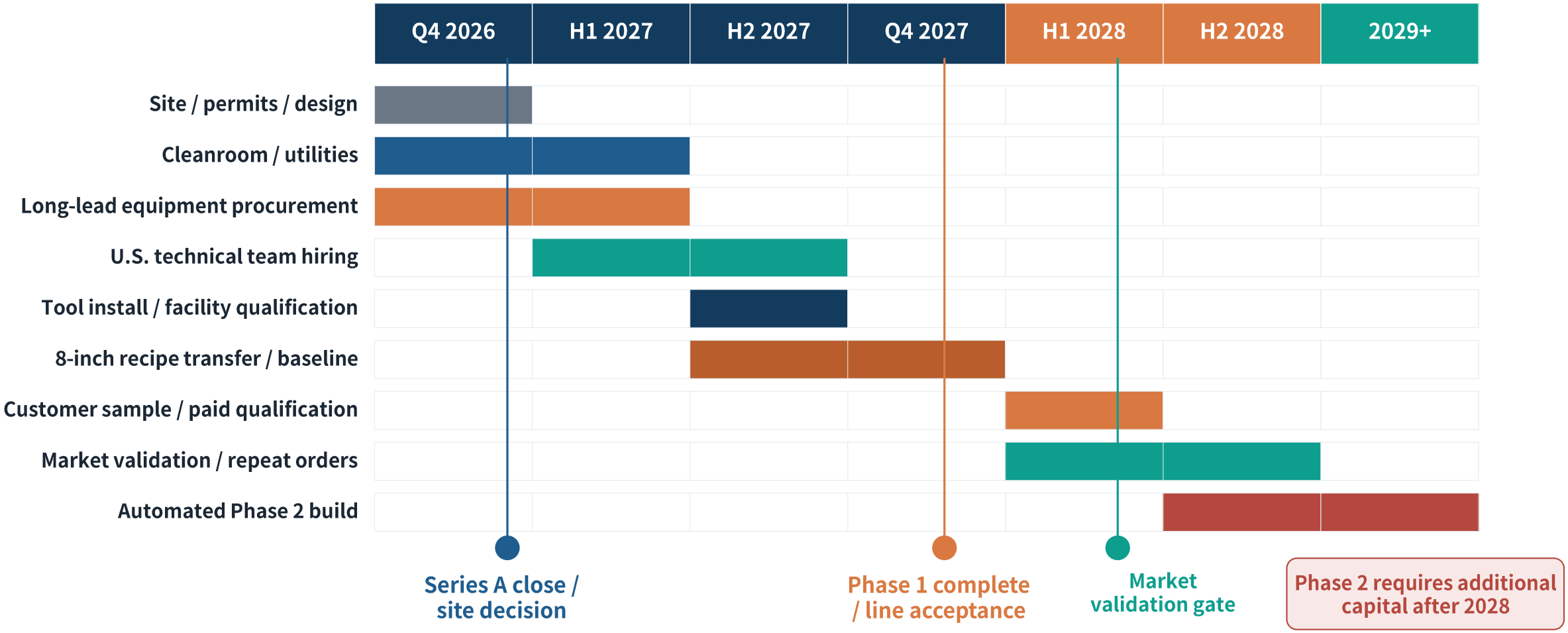
- Packaging and fiber coupling
- Cryogenic device test
- Advanced SEM / AFM / optical metrology
- University and national-lab programs
- Scale-foundry and strategic partners

**Phase 1: customer qualification platform (2026-2028)**
**Phase 2: automated 8-inch quantum photonics foundry (post-2028, additional capital)**

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# U.S. Expansion – Timeline

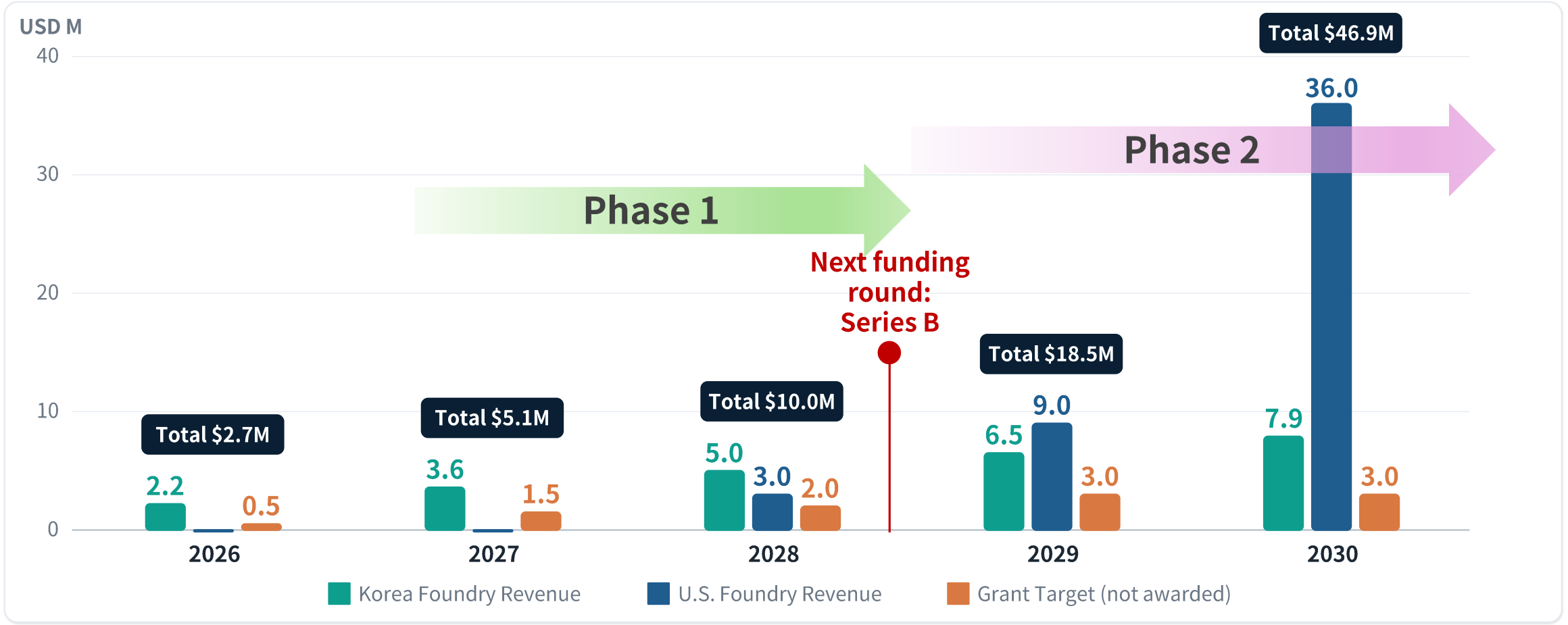
> Cleanroom work: late 2026, Phase 1: completed by end 2027, Customer & Market Validation: early 2028.




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# Financial Projections

> Model growth from $2.7M in 2026 to $46.9M in 2030, U.S. foundry as main growth driver after 2028.




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# Vision & Ask

> Funds focused on U.S. Phase 1 to convert KR-developed materials & process IP into 8-inch U.S. manufacturing platform.

**Target:**
**$8M***
Series A
Pre-money: 15-20%**

## Use of Funds: U.S. Foundry

- Foundry Phase 1
- 8-inch + nano-patterning → PICs
- Team Expansion
- International Events & Marketing

- * + Projected $2M USD/year as non-dilutive funding (Korea & US R&D grants)
- ** Open to discussion

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# Management Team

> PhD materials leadership, thin-film deposition research team, IP & cleanroom construction expertise



## Dr. Soodeok Han

**CEO / Founder**

PhD in Engineering; semiconductor thin films; commercialization and foundry strategy.



## Dr. Wonyoung Choi

**CTO**

PhD in Engineering; semiconductor thin films; process and R&D leadership.



## Dr. Sahar Ayachi

**CSO**

PhD in Engineering; semiconductor devices; international technical programs.



## Dr. Hyeonkyo Song

**Principal Researcher**

PhD in Physics; semiconductor devices and advanced materials research.

**PLANNED NEXT-STAGE HIRE**

## Finance Lead

- U.S. subsidiary finance
- Grant accounting
- Capex control
- Investor reporting
- Follow-on fundraising

**OPERATING CAPABILITIES**

| 13 | PhD-led | Deposition | IP | Cleanroom |
|---|---|---|---|---|
| Employees (full-time) | Founding & Management team | Specialist research staff | Patent attorney expertise | Infrastructure architect |

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| CSO
| Contact

Dr Sahar Ayachi
Ayachi.sahar@vanam.co.kr
www.vanam.co.kr

VANA MATERIALS LAB.



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# Technology: AI-Assisted Process Platform: The Recipe-Transfer and Yield Engine

**In-situ sensing, multivariate statistical monitoring and closed-loop diagnostics convert operator know-how into transferable digital recipes — directly de-risking the Korea-to-U.S. 8-inch transfer.**

**CLOSED-LOOP PROCESS PLATFORM (in production use, Seoul foundry)**

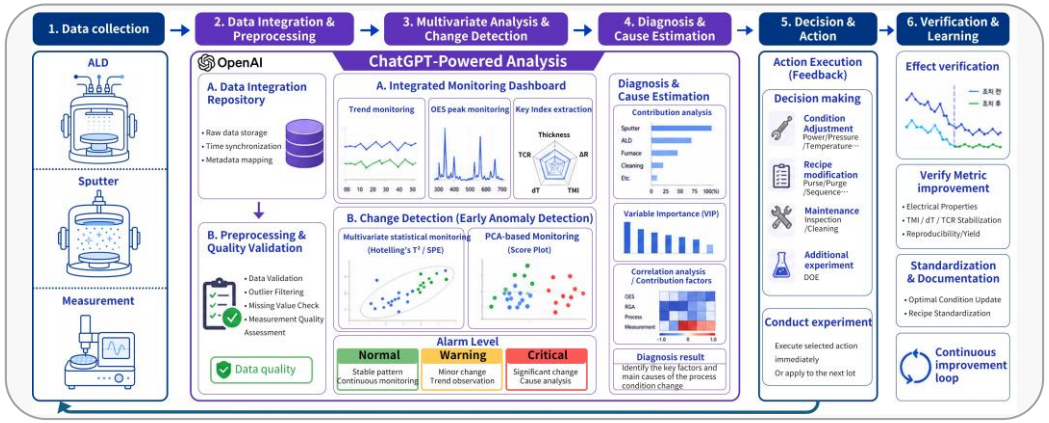
1. **Data Collection**
   ALD / sputter tool signals and in-situ sensing (incl. OES) plus film metrology, logged per run.
2. **Integration & Preprocessing**
   Time-synchronized repository, metadata mapping, outlier and quality validation.
3. **Multivariate Monitoring**
   Hotelling's $T^2$ / SPE control and PCA score monitoring detect drift before scrap is produced.
4. **Diagnosis**
   Contribution and variable-importance (VIP) analysis isolates the root-cause parameter.
5. **Action**
   Condition adjustment, recipe modification, maintenance trigger or targeted DOE.
6. **Verification & Standardization**
   Effect confirmed on metrics (reproducibility, yield); recipe updated and versioned.

↺ **Loop closes: verified recipe becomes the new baseline for the next run**

**Transfer risk down**
A recipe is a versioned data object with its sensor fingerprint, not tribal knowledge. The U.S. 8-inch line inherits the same monitoring stack from day one; cross-site golden samples close the loop.

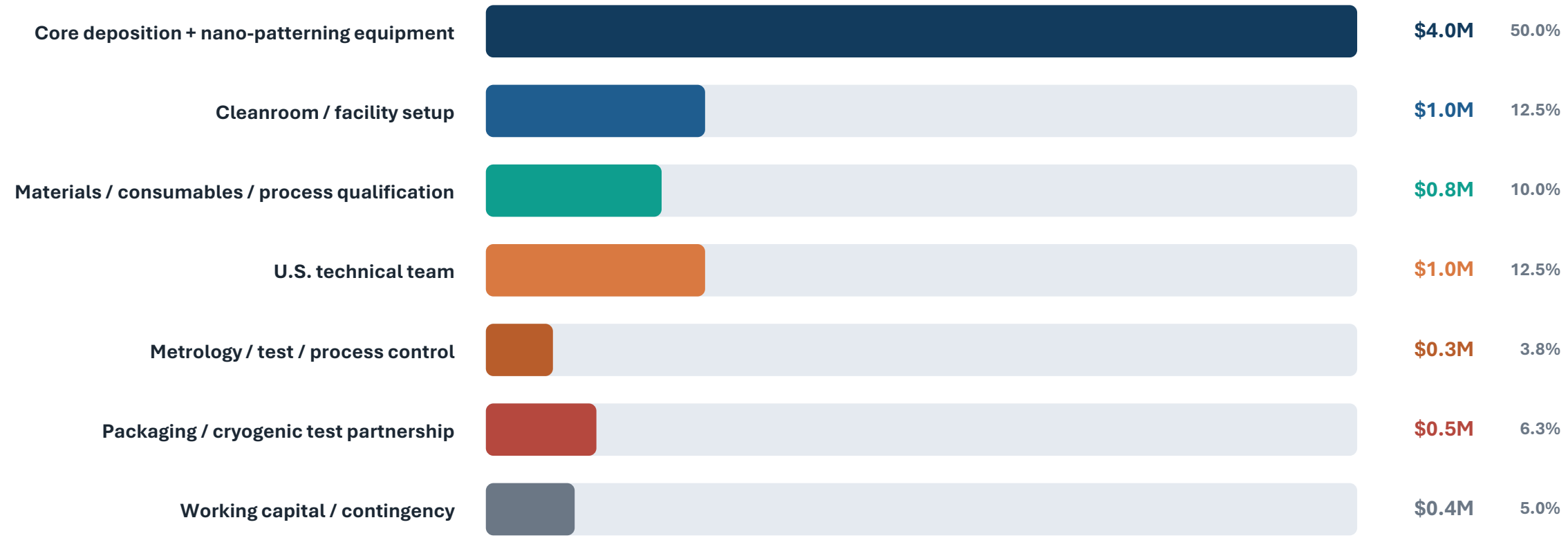
**Yield economics**
Early anomaly detection reduces scrapped runs and shortens qualification DOE cycles; in a low-volume, high-ASP model every avoided scrap run is direct gross margin.

**Compounding moat**
Every customer run adds labeled process data; the six-layer moat (patents, recipes, process windows, metrology correlations — see Defensibility) becomes machine-readable and auditable in diligence.



# Use of Proceeds: $8.0M Fully Allocated to U.S. Phase 1

**Half of the round funds core deposition and nano-patterning equipment; internal construction capability and Korea metrology leverage keep facility and test budgets disciplined.**




**TOTAL SERIES A USE: $8.0M**

**Facility rationale** Internal cleanroom design and construction-management capability is expected to reduce setup cost.

**Metrology rationale** Korea equipment and external partners reduce the need to purchase every advanced metrology tool in Phase 1.

# Exit Strategy: Revenue Scale or IP-Driven Strategic Value

**VanaM has multiple credible exit pathways because strategic value can reside in qualified foundry capacity, thin-film process IP, material recipes and nano-patterned quantum-device IP.**

**VanaM**

## Strategic Asset Base

Thin-film process IP
Material recipes
Qualified foundry capability

### 1 IPO

Specialty semiconductor, photonics or quantum supply-chain company.

### 2 Quantum OEM M&A

Vertical integration of critical photonic materials and device capability.

### 3 Foundry M&A

Photonics foundry or specialty semiconductor foundry expansion.

### 4 Materials M&A

Advanced materials company seeking device-level integration.

### 5 IP Acquisition / Licensing

Monetization of recipes, process modules and device IP.

### 6 JV / Strategic Investment

National-security or quantum-infrastructure players.

**Exit readiness = clean IP ownership + qualified processes + repeat customers + transferable know-how + credible U.S. manufacturing.**

APPENDIX

# Three-Year Financial Projection: U.S. Series A Funding Plan

## H2 2026–H1 2029, USD M; Series A outflow totals match the $8.0M U.S. Phase 1 use-of-proceeds budget.

### CAPITAL OUTFLOW — U.S. PHASE 1 USE OF PROCEEDS

| Capital Outflow ($M) | 2026 H2 | 2027 H1 | 2027 H2 | 2028 H1 | 2028 H2 | 2029 H1 | Total |
|---|---|---|---|---|---|---|---|
| Equipment cost | 1.2 | 2.2 | 0.6 | 0.0 | 0.0 | 0.0 | **4.0** |
| Facility cost | 0.5 | 0.4 | 0.1 | 0.0 | 0.0 | 0.0 | **1.0** |
| Material & consumable cost | 0.1 | 0.1 | 0.1 | 0.2 | 0.2 | 0.1 | **0.8** |
| Technical team cost | 0.1 | 0.2 | 0.2 | 0.2 | 0.2 | 0.1 | **1.0** |
| Other cost | 0.1 | 0.1 | 0.2 | 0.3 | 0.3 | 0.2 | **1.2** |
| **TOTAL OUTFLOW** | **2.0** | **3.0** | **1.2** | **0.7** | **0.7** | **0.4** | **8.0** |

### CAPITAL INFLOW — REVENUE, INVESTMENT AND GRANT TARGETS

| Capital Inflow ($M) | 2026 H2 | 2027 H1 | 2027 H2 | 2028 H1 | 2028 H2 | 2029 H1 | Total |
|---|---|---|---|---|---|---|---|
| U.S. foundry revenue | 0.0 | 0.0 | 0.0 | 1.0 | 2.0 | 4.5 | 7.5 |
| Series A investment | 8.0 | 0.0 | 0.0 | 0.0 | 0.0 | 0.0 | 8.0 |
| Grant target (not awarded) | 0.5 | 0.5 | 1.0 | 1.0 | 1.0 | 1.5 | 5.5 |
| **TOTAL INFLOW** | **8.5** | **0.5** | **1.0** | **2.0** | **3.0** | **6.0** | **21.0** |

### OUTFLOW ITEM DEFINITIONS

**Equipment:** 8-inch deposition, lithography/direct-write, RIE/ICP-RIE, lift-off/clean and acceptance.
**Facility:** site/design, cleanroom, utilities, exhaust, permitting and code compliance.
**Materials:** substrates, targets, gases, resist/chemicals and DOE/qualification lots.
**Technical team:** core U.S. process, equipment and operations roles; exact headcount TBD.
**Other:** metrology/test process control, packaging/cryogenic-test partners and working-capital contingency.

### MODEL BOUNDARY / VERIFICATION

Revenue row uses U.S. foundry revenue only; Korea operating revenue is excluded. **Grant targets are not awarded.** Revenue and grant targets are planning inputs, not cash runway or profitability. H2 2029 is outside this three-year window.

# Non-Dillutive Capital & Grant Strategy: $10M Target, Not Yet Awarded

**VanaM is discussing a CHIPS Act-related proposal with Intelon Optics, but the five-year grant schedule remains a proposal target and must not be treated as committed funding.**

## CURRENT STATUS

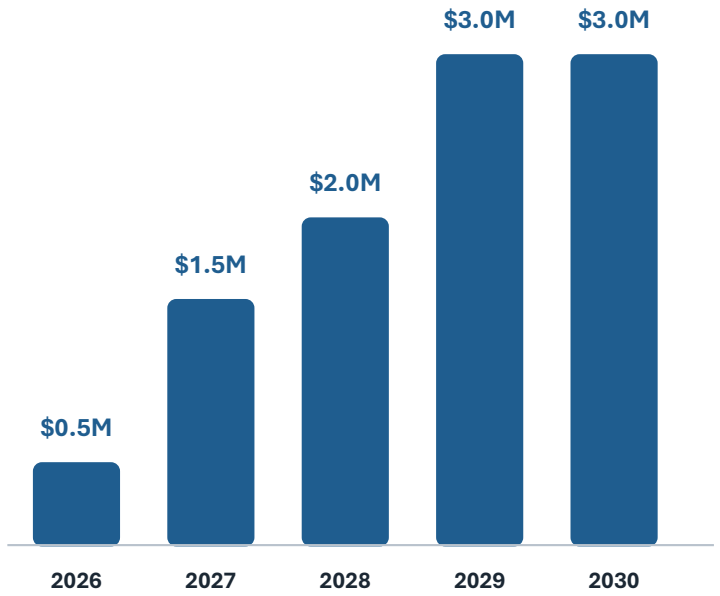
**Proposal under discussion**

NOT YET AWARDED

- Potential CHIPS Act-related project with Intelon Optics.
- No grant agreement, agency award or committed disbursement.
- Should be excluded from liquidity until formally awarded.
- Treat as a financing upside and validation pathway.

**$10.0M five-year target**

## TARGET SCHEDULE




Proposal target only

## READINESS GATES

1. U.S. applicant and eligibility
2. Eligible cost and cost-share map
3. Work packages and measurable milestones
4. IP, data and publication terms
5. **Audit-ready grant accounting**
6. **Capex controls and investor reporting**

**Next-stage U.S. finance lead owns gates 5-6.**

# Operations: Korea Production Capacity + U.S. Integrated PIC Revenue Potential

**Capacity scenarios combine the current Korea materials / parts flow with a downstream U.S. device foundry that patterns, etches, tests and packages integrated quantum photonic PICs.**




**<U.S. Foundry Integrated PIC Estimate>**

- ✓ Korea material bottleneck: STO / DBR ~30 good wafers per month × 20 released 1Qubit-eq PIC die per wafer = 600 bare PIC die per month
- ✓ Final packaged sales case: 50% package / cryogenic qualification release from bare die = 300 units per month
- ✓ **ASP basis: USD 13K bare die midpoint; USD 30K packaged 1Qubit-equivalent PIC**

| Product | Sales Basis (USD) | Monthly Units | Monthly Revenue (USD) | IR Interpretation |
|---|---|---|---|---|
| **KOREA FOUNDRY: MATERIALS / COMPONENT CAPACITY SCENARIOS** | | | | |
| $Ta_2O_5$ photonics wafer, 4-inch | USD 893/wafer | 1,296 wafers | **USD 1,157K** | Volume-oriented capacity anchor |
| STO & BTO substrate wafer, 4-inch | USD 2,500/wafer | 30 wafers | **USD 75K** | High-spec, limited-throughput modulator material |
| DBR mirror, 4-inch | USD 13,214/unit | 30 units | **USD 396K** | High-value multilayer mirror product |
| SNSPD base wafer, 3-inch | USD 5,714/wafer | 64 wafers | **USD 346K** | Most direct SNSPD wafer model |
| SNSPD chip die | USD 2,357/die | 896 die | **USD 2,112K** | Higher-value model after pattern / screening |
| Packaged SNSPD chip | USD 9,643/unit | 100 units | **USD 964K** | Capped by package / cryogenic test |
| **U.S. FOUNDRY: INTEGRATED QUANTUM PHOTONIC PIC FINAL STAGE** | | | | |
| **Integrated Quantum PIC die (1 Qubit-eq, U.S.)** | USD 13,200/die(base ASP) | 600 die | **USD 7,920K** | Bare die after U.S. lithography / etch / release; USD 6.4M-9.4M range |
| **Packaged Quantum PIC (1 Qubit-eq, U.S.)** | USD 30,000/unit(base ASP) | 300 units | **USD 9,000K** | Final low-volume product after fiber / electrical package & cryogenic qualification |